Specify the force. Guide the load. Control the stroke.

Water hydraulic cylinders turn pressure into linear movement. Select bore, rod, stroke and mounting together, and provide external guidance for side loads. Push force and pull force are different design checks.

At the same pressure, the rod reduces the effective area on retraction. Check push and pull loads separately, including return pressure and friction. Use the force tables as an ideal sizing reference and confirm the component rating for the chosen construction.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Model selector

Product / configuration	Published selection information
100 mm (3.94 in) bore × 60 mm (2.36 in) rod: cylinder force data	Piston Area 78.54 cm² (12.174 in²); Ring Area 50.27 cm² (7.792 in²)
100 mm (3.94 in) bore × 80 mm (3.15 in) rod: cylinder force data	Piston Area 78.54 cm² (12.174 in²); Ring Area 28.27 cm² (4.382 in²)
125 mm (4.92 in) bore × 80 mm (3.15 in) rod: cylinder force data	Piston Area 122.72 cm² (19.022 in²); Ring Area 72.45 cm² (11.230 in²)
32 mm (1.26 in) bore × 16 mm (0.63 in) rod: cylinder force data	Piston Area 8.04 cm² (1.246 in²); Ring Area 6.03 cm² (0.935 in²)
32 mm (1.26 in) bore × 20 mm (0.787 in) rod: cylinder force data	Piston Area 8.04 cm² (1.246 in²); Ring Area 4.9 cm² (0.760 in²)
40 mm (1.57 in) bore × 25 mm (0.984 in) rod: cylinder force data	Piston Area 12.57 cm² (1.948 in²); Ring Area 7.66 cm² (1.187 in²)
50 mm (1.97 in) bore × 25 mm (0.984 in) rod: cylinder force data	Piston Area 19.63 cm² (3.043 in²); Ring Area 14.73 cm² (2.283 in²)
50 mm (1.97 in) bore × 30 mm (1.18 in) rod: cylinder force data	Piston Area 19.63 cm² (3.043 in²); Ring Area 12.57 cm² (1.948 in²)
64 mm (2.52 in) bore × 30 mm (1.18 in) rod: cylinder force data	Piston Area 32.17 cm² (4.986 in²); Ring Area 25.1 cm² (3.891 in²)
64 mm (2.52 in) bore × 40 mm (1.57 in) rod: cylinder force data	Piston Area 32.17 cm² (4.986 in²); Ring Area 19.6 cm² (3.038 in²)
80 mm (3.15 in) bore × 50 mm (1.97 in) rod: cylinder force data	Piston Area 50.27 cm² (7.792 in²); Ring Area 30.63 cm² (4.748 in²)

Specifications and selection data

Bore Ø	Rod Ø	Piston Area	Ring Area	Force Extending @ 160 bar (2,321 psi)
32 mm (1.26 in)	16 mm (0.63 in)	8.04 cm² (1.246 in²)	6.03 cm² (0.935 in²)	12.61 kN (2,835 lbf)
32 mm (1.26 in)	20 mm (0.787 in)	8.04 cm² (1.246 in²)	4.9 cm² (0.760 in²)	12.61 kN (2,835 lbf)
40 mm (1.57 in)	25 mm (0.984 in)	12.57 cm² (1.948 in²)	7.66 cm² (1.187 in²)	19.70 kN (4,429 lbf)
50 mm (1.97 in)	25 mm (0.984 in)	19.63 cm² (3.043 in²)	14.73 cm² (2.283 in²)	30.79 kN (6,922 lbf)
50 mm (1.97 in)	30 mm (1.18 in)	19.63 cm² (3.043 in²)	12.57 cm² (1.948 in²)	30.79 kN (6,922 lbf)
64 mm (2.52 in)	30 mm (1.18 in)	32.17 cm² (4.986 in²)	25.1 cm² (3.891 in²)	50.44 kN (11,339 lbf)
64 mm (2.52 in)	40 mm (1.57 in)	32.17 cm² (4.986 in²)	19.6 cm² (3.038 in²)	50.44 kN (11,339 lbf)
80 mm (3.15 in)	50 mm (1.97 in)	50.27 cm² (7.792 in²)	30.63 cm² (4.748 in²)	78.82 kN (17,719 lbf)
100 mm (3.94 in)	60 mm (2.36 in)	78.54 cm² (12.174 in²)	50.27 cm² (7.792 in²)	123.15 kN (27,685 lbf)
100 mm (3.94 in)	80 mm (3.15 in)	78.54 cm² (12.174 in²)	28.27 cm² (4.382 in²)	123.15 kN (27,685 lbf)
125 mm (4.92 in)	80 mm (3.15 in)	122.72 cm² (19.022 in²)	72.45 cm² (11.230 in²)	192.42 kN (43,258 lbf)

Bore Ø	Force Retracting @ 160 bar (2,321 psi)	Max Pressure	Efficiency
32 mm (1.26 in)	9.46 kN (2,127 lbf)	160 bar (2,321 psi)	0.98
32 mm (1.26 in)	7.68 kN (1,727 lbf)	160 bar (2,321 psi)	0.98
40 mm (1.57 in)	12.01 kN (2,700 lbf)	160 bar (2,321 psi)	0.98
50 mm (1.97 in)	23.09 kN (5,191 lbf)	160 bar (2,321 psi)	0.98
50 mm (1.97 in)	19.70 kN (4,429 lbf)	160 bar (2,321 psi)	0.98
64 mm (2.52 in)	39.36 kN (8,848 lbf)	160 bar (2,321 psi)	0.98
64 mm (2.52 in)	30.74 kN (6,911 lbf)	160 bar (2,321 psi)	0.98
80 mm (3.15 in)	48.03 kN (10,798 lbf)	160 bar (2,321 psi)	0.98
100 mm (3.94 in)	78.82 kN (17,719 lbf)	160 bar (2,321 psi)	0.98
100 mm (3.94 in)	44.33 kN (9,966 lbf)	160 bar (2,321 psi)	0.98
125 mm (4.92 in)	113.61 kN (25,541 lbf)	160 bar (2,321 psi)	0.98